Work instructions for Engine inspection and ECU software update Service Campaign – Service Campaign 020219

About this work

<Issue>
If the vehicle is not refuelled with at least 15 litres of fuel in a single fill-up within approximately three months, a message should be displayed prompting the driver to refuel with 15 litres or more. Additionally, the engine should start automatically. However, it has been found that this function does not operate as intended due to an issue in the engine ECU's control program.

<Engine Inspection Procedure>
To assess any potential impact on the engine and fuel system components caused by fuel deterioration, an engine inspection shall be conducted.
The inspection must be conducted in **Battery Charge Mode**. **Customers are to be asked at the time of the service booking to bring their vehicle in with the drive battery level below 50%.**

Important:
The use of Idle Continuation Mode is not permitted, as it does not apply sufficient engine load to allow for an accurate inspection.
Ensure that all inspections are strictly carried out in **Battery Charge Mode**.

<Engine ECU Update>
To ensure proper operation of the control logic that prompts the driver to refuel with at least 15 litres and automatically starts the engine if this condition is not met within approximately 3 months, the engine ECU will be updated. Without this update, the control functions may not activate as intended.

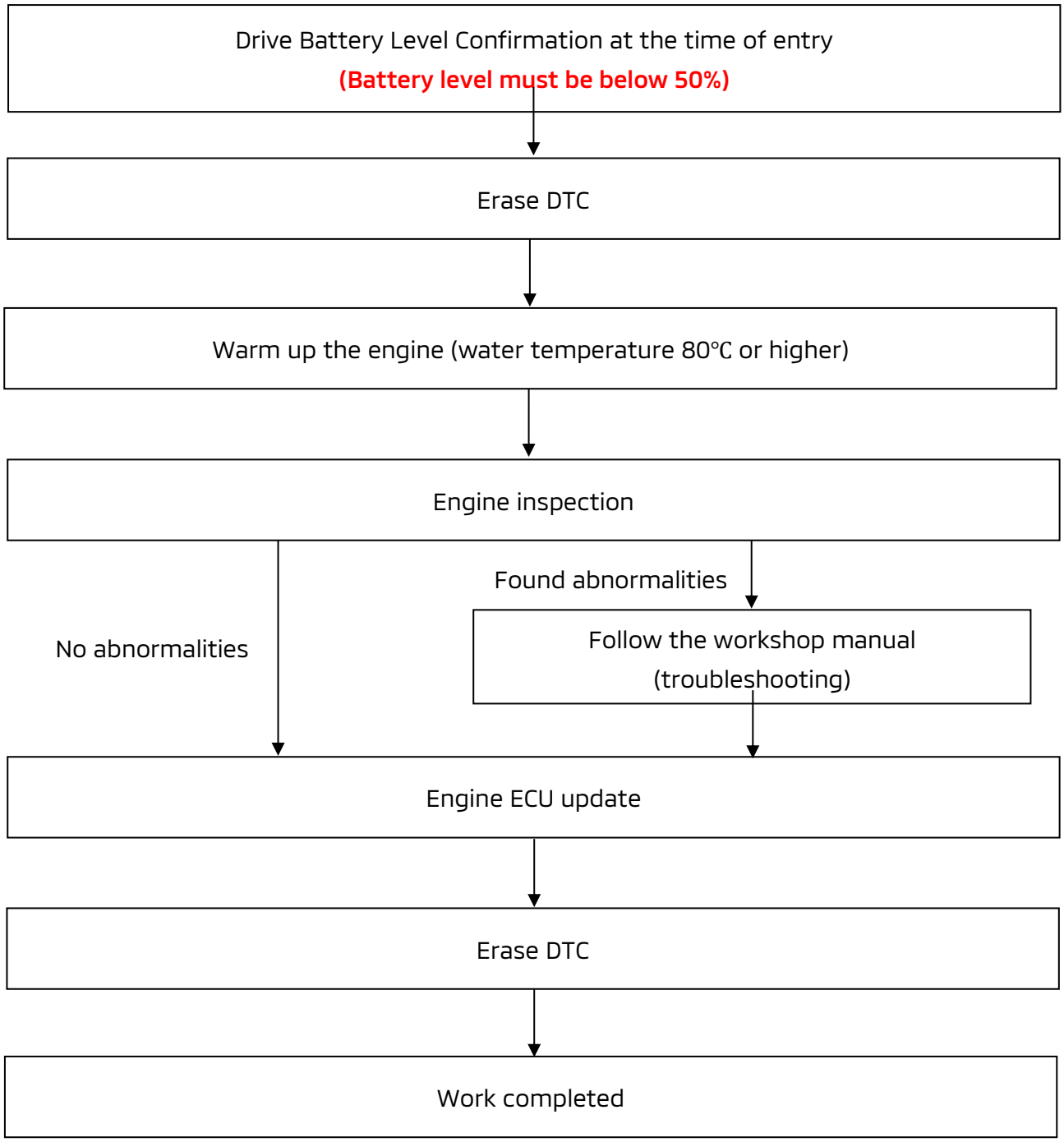
<Fuel Refill Monitoring After ECU Update>
The countdown for the elapsed days begins from the point at which the ECU update is completed.

Drive Battery Level Confirmation at the time of entry

- **At the time of vehicle entry, confirm that the drive battery level is below 50%.**
  If the battery level exceeds 50%, there is a risk that Battery Charge Mode may either stop during the engine inspection or fail to activate.

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020219 Work flow



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0. Check the Drive Battery Level at the time of entry

When the vehicle arrives, make sure that the Drive Battery Level is less than 50%. If it is more than half, the Battery Charge Mode may stop during engine inspection or may not be able to enter Battery Charge Mode.

1. Erase DTC

(1.1) Connect the MUT-IIISE and erase the DTCs by selecting "ALL DTC" ⇒ "All Diagnosis (Erase & Read)" and confirm that no DTCs appear.

☞Perform this in "READY" mode

☞Erase all DTCs once to check for fuel system abnormalities (lean) and cylinder misfire detection system DTCs as listed in "3. Engine Inspection"

*If a DTC appears, take action based on the warranty period, just as with general maintenance/inspections.

⚠ Caution

- Turn off the vehicle's air conditioning, seat heaters, audio, etc. to reduce power consumption.
- Work indoors to prevent the PC from freezing or shutting down due to abnormal heat generation.
- Use an external AC power source for the PC using MUT-IIISE to prevent the battery from running out.

2. Warm-up the engine (water temperature 80°C or higher)

(2.1) Open the engine hood and warm up the engine (water temperature 80°C or higher).

☞Start the engine in **Battery Charge Mode** with the MUT-IIISE connected.

☞Leave the engine running until the service data item No. 6 Water Temperature Sensor reaches 80°C or higher.

⚠ Caution

- Do not leave the vehicle unattended. If the vehicle is left unattended for a long time, the remaining drive battery charge may increase, causing the Battery Charge Mode to stop or making it impossible to enter Battery Charge Mode.
- The "continuous idle mode" is not permitted as the engine load is low and proper inspection cannot be performed.
- Always carry out the inspection in **Battery Charge Mode**

(2.2) Once the service data item No. 6 Water Temperature Sensor reaches 80°C or above, continue idling for 10 minutes in **Battery Charge Mode**.

☞To check the DTC and fuel correction value

⚠ Caution

- Check that the intake air temperature sensor 1 (service data item No. 5) on the MUT-IIISE is below 50°C. If it is above 50°C, use a fan or similar to blow air into the engine compartment (around the intake duct above the left hand light).
- ☞If the intake air temperature is above 50°C, the inspection may not be performed properly.
- If the vehicle is left for a long time, the remaining drive battery charge may increase, causing the Battery Charge Mode to stop during engine inspection or making it impossible to enter Battery Charge Mode. Do not leave the vehicle unattended for more than 10 minutes.
- The Continuous Idle Mode is not permitted as the engine load is low and a proper inspection cannot be performed.
- **Be sure to perform the inspection in Battery Charge Mode**

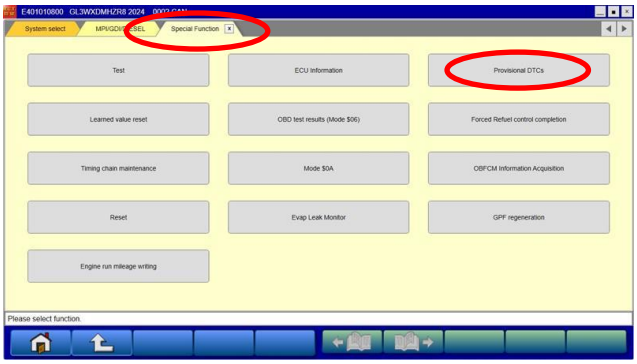
3. Engine inspection (DTC check)

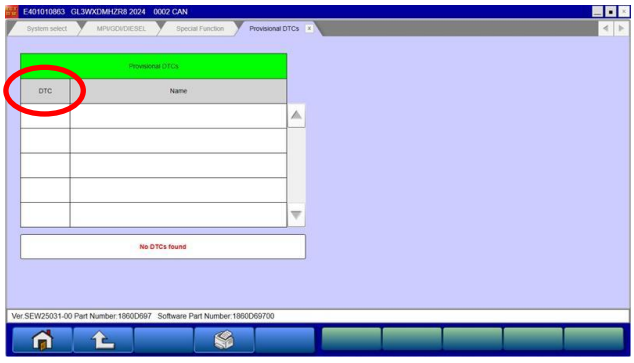
(3.1) While in **Battery Charge Mode with engine running**, check whether the following DTCs appear to the engine ECU due to fuel deterioration:

P0171: Fuel system abnormality (lean)
P0300: Multi-cylinder misfire detection system
P0301: No. 1 cylinder misfire detection system
P0302: No. 2 cylinder misfire detection system
P0303: No. 3 cylinder misfire detection system
P0304: No. 4 cylinder misfire detection system

Check whether a provisional DTC appears from "MPI/GDI/Diesel" ⇒ "Special Function" ⇒ "Provisional DTCs" on the MUT-IIISE.

If a provisional DTC is displayed, follow the troubleshooting instructions in the workshop manual (Refer to the SSI when applying for troubleshooting labour times, part replacement labour times, and part costs).





⚠ Advice

Checking the provisional fault code

- The engine ECU will temporarily store some DTCs if a fault is detected during the first drive cycle (while the power switch power mode is ON), and if the same fault is detected during the next drive cycle, it will be confirmed as a fault, the DTC will be stored and the engine warning light will be turned on.
- The MUT-IIISE can display the provisional fault code stored by the engine ECU when detected during the first drive cycle.
- This makes it possible to check in a single drive cycle whether the fault will reoccur after repairs.

(3.2) While in **Battery Charge Mode**, check the MUT-IIISE service items.
Check that the engine ECU service data items in the table below are not NG criteria.
("MPI/GDI/Diesel" ⇒ "Service Data")

Table Engine ECU Service Data

	Item No.	Inspection Content	NG Criteria
Long Time Fuel Correction Value 1	26	Correction Value	**+10% or more** (Check to see if it is exceeded on the increase side)
Real Time Fuel Correction Value 1	28	Correction Value	**＋25% or more** (Check to see if it is exceeded on the increase side)

If the NG criteria are met, follow the inspection procedure for troubleshooting DTC "**P0171**" in the workshop manual.

(3.3) Turn the power switch power mode to OFF.

4. Engine ECU update

(4.1) Connect the auxiliary battery to external power

⚠ Caution

Battery Charger Connection During ECU Update

- During the ECU update process, always connect a battery charger and ensure backup power is supplied to the auxiliary battery.
- If the auxiliary battery has low capacity, the ECU update may fail due to battery depletion. This can result in corruption of the ECU control program, potentially requiring replacement with a new ECU.
- To prevent such issues, confirm that the battery charger is properly connected and functioning before starting the update.

(4.2) Turn the power switch power mode to ON.

☞ Press the power switch twice without depressing the brake pedal.

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(4.3) Update the engine ECU.
☞ Update time is approximately 2 minutes.

⚠ Caution

Precautions for ECU Update Procedure

Prevent Sleep Mode
Configure the PC settings to prevent it from entering sleep mode during the update process (Screen lock is acceptable).
If the PC enters sleep mode during the update, the process may terminate prematurely, potentially requiring replacement of the ECU with a new unit.

Maintain Network Connection
The ECU update must be performed in an environment with a stable internet connection.
Ensure that the PC remains properly connected to the network throughout the update.

Vehicle and ECU Verification (MUT-III SE)
Before proceeding with the update using MUT-III SE, confirm that the vehicle is the correct target for the update.
Carefully verify and select the correct ECU to avoid applying the update to the wrong unit.

Update Type Selection
If the update type selection screen appears in MUT-III SE, select:
“Reprogramming (ECU Software Rewrite)”

Data Verification
Confirm that the downloaded data matches the specifications listed in this procedure manual.

Post-Update Confirmation
After the update is complete, verify that the written data matches the data specified in this procedure manual.
Once confirmed, press the OK button to finalize the process.

(4.4) After updating, confirm that the **New SW Part No/Ver** is "**1860D69701**"
("MPI/GDI/Diesel" ⇒ "Special Functions" ⇒ "ECU Information")

5. Erase DTC

(5.1) DTC will appear when the engine ECU is updated. Erase it using "All diagnostics (erase & read)."
☞To mark the DTC as a past fault, turn the power switch power mode OFF once, then ON again before erasing the DTC.

WORK COMPLETED